YUVA YODHA ENERGY TECH HACKATHON 2026 · CHALLENGE 01

CropClock

Scheduling India's farm power to the crop, not the calendar.

Sustainable Agriculture — Energy, Water & Productivity Case study: Maharashtra, MSKVY feeders

A fifth of India's electricity is scheduled by a roster written months in advance.

21.2 M

grid-electrified agricultural pumpsets

CEA, 2019

~1 / 5

of national electricity consumption goes to agriculture

CEEW · ICED-NITI

8 h

daily supply window per feeder, fixed by rota

MSEDCL agricultural policy

- The roster is blind to **rainfall** — power arrives the morning after 40 mm of rain.
- It is blind to **crop growth stage** — every week is treated identically, though water demand varies several-fold.
- It is blind to **the feeder's own solar** — pumps run at 06:00 while the plant peaks at noon.

Farmers over-irrigate because supply is unpredictable — not because they are uninformed.

If you cannot forecast when power returns, flooding the field while you have it is the rational choice. Every existing product targets the farmer's *knowledge*. None targets the *uncertainty* that actually drives the behaviour.

> **Reliability is itself a water-saving technology.**

Telangana proves it: 24×7 free agricultural supply since January 2018 removed the uncertainty and changed behaviour — at a fiscal cost its DISCOMs cannot carry indefinitely. **CropClock delivers the predictability without the kWh.**

Two mature fields that have never been connected.

SATELLITE CROP WATER STRESS

SEBAL · SSEBop · METRIC

Fasal · Fyllo · Farmonaut

Advises **the farmer**

Assumes power is already available

AGRICULTURAL FEEDER SCHEDULING

Jyotigram, 2003 → 6+ states

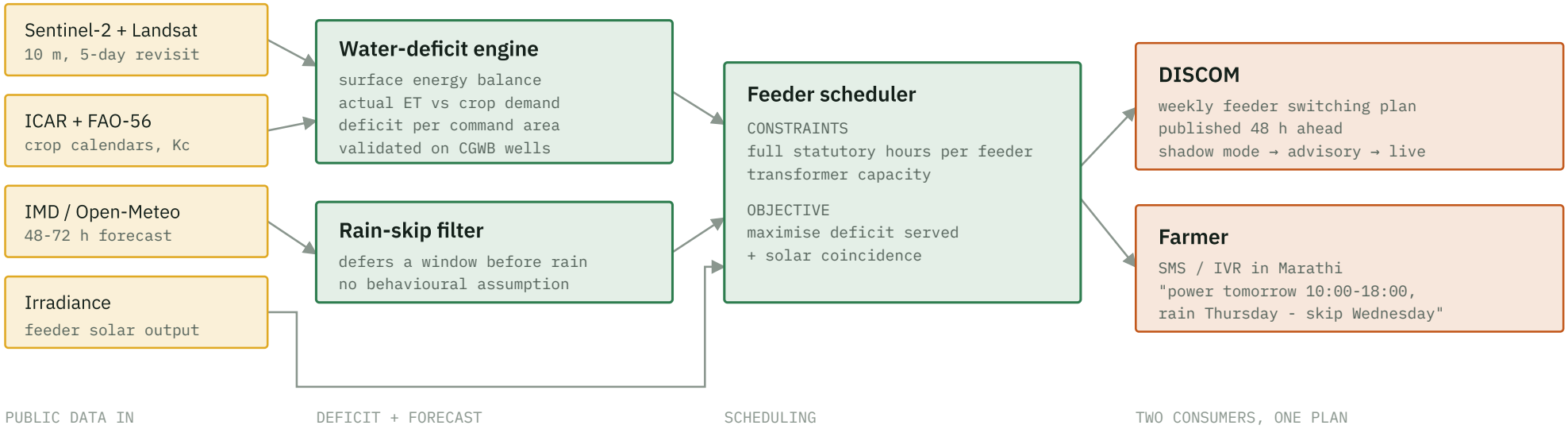
MSEDCL 8-hour daytime rota

Optimises **fairness and load**

Has never once looked at a crop

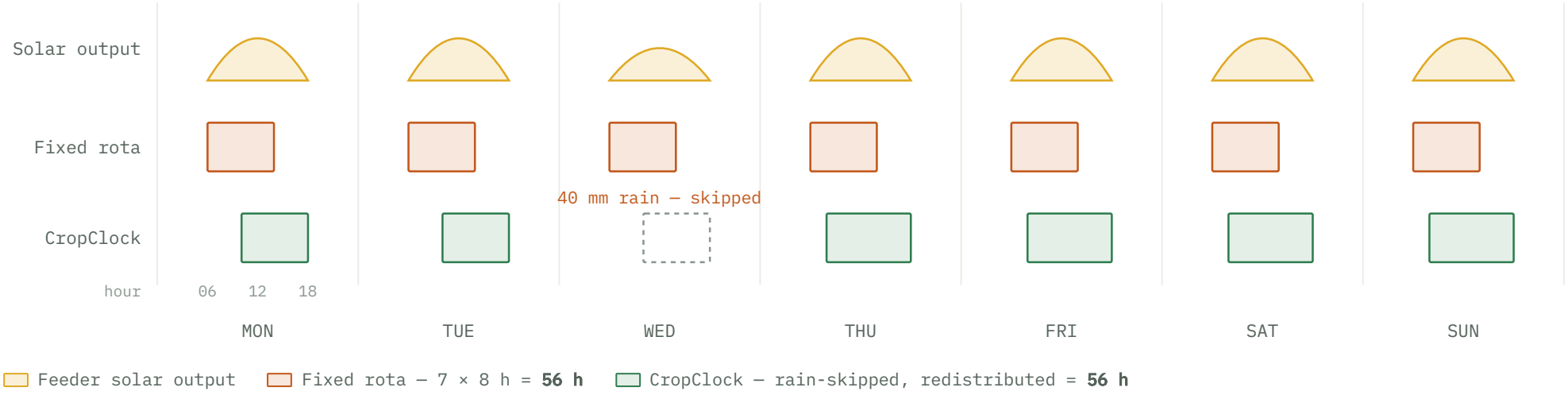
> **CropClock is the missing link between them.**

Four components, one weekly decision per feeder.



The farmer message goes out in Marathi — फीडर ४७: उद्या सकाळी १०:०० ते संध्याकाळी ६:०० वीजपुरवठा. Every feeder still receives its **full statutory hours**: the saving is timing, not rationing.

Same 56 hours. Delivered when the crop and the sun agree.



WHAT CHANGED ON WEDNESDAY

40 mm of rain is forecast, so the window is **cancelled outright**.

Those 8 hours are redistributed across Thursday to Sunday, +2 h each.

The feeder's weekly entitlement is untouched. Nobody is rationed.

WHAT CHANGED EVERY OTHER DAY

Windows shift from **06:00–14:00** to **08:00–18:00**.

Pumping now sits under the feeder's own solar generation curve.

The DISCOM buys fewer units from the grid to serve the same irrigation.

From reflectance to a switching decision.

- **Actual ET** — surface energy balance, SSEBop-style, on Sentinel-2 (10 m, 5-day revisit) with Landsat thermal.
- **Crop demand** — FAO-56 reference ET × crop coefficient, staged against ICAR crop calendars.
- **Deficit** — demand − (actual ET + effective rainfall), as a running soil-water balance per command area.
- **Scheduler** — fairness-constrained window assignment under transformer capacity; greedy heuristic first, MILP if time allows.
- **Validation** — CGWB observation-well levels as an independent check on seasonal groundwater draft.

Every input is public and free. Sentinel-2, Landsat, IMD, CGWB, MERC tariff orders, Grid-India. No sensors, no field deployment, no proprietary data — which also means anyone can reproduce our numbers.

One feeder, one year of observed weather: 128 MWh and ₹9.6 lakh.

35.2 → 48.6%

of pump energy served by the feeder's own solar plant, rabi

+13.4 points

128.2 MWh

grid energy avoided per feeder per year

9.1% of baseline pumping

₹9.62 L

subsidy avoided per feeder per year

at ₹7.50 / kWh cost of supply

Solapur district, seasons of 2025–26. Observed rainfall, reference ET and hourly irradiance from the ERA5 archive — 250 pumps of 7.5 HP on a 50 m borewell, 500 ha, 1 MW MSKVY plant. **Tier A only: no behavioural assumption enters this figure.** Against the 2,269 MW already tariff-approved under MSKVY 2.0, this scales to roughly **291 GWh and ₹218 crore a year**.

We led with rain-skip. The data said solar alignment.

RABI — OCT TO MAR

101 mm rain against 839 mm reference ET

Zero irrigations worth deferring. In the dry season there is no rain to skip.

Solar alignment is the entire Tier A saving: **106.7 MWh**.

KHARIF — JUN TO OCT

1,112 mm rain against 657 mm reference ET

Only one crop-optimal irrigation all season — but two deferrals against realistic behaviour.

Rain-skip saves **255.5 MWh**, more than the whole rabi solar figure.

> **The lever is different in each season. A single-season pilot draws the wrong conclusion.**

And the largest number in the whole model is neither: defensive irrigation applies **1,200 mm** in kharif where 138 mm meets crop demand, and **1,092 mm** of that drains past the root zone. Closing that gap is what the published-in-advance schedule is actually for.

Software on infrastructure India has already funded.

16,000 MW

MSKVY 2.0 feeder-solarisation target

Maharashtra GR, 13.09.2024

2,269 MW

tariffs approved under MSKVY 2.0

MERC, August 2026

₹0

new hardware capex required to deploy

public data only

FARMER

Predictable, adequate irrigation. Less wasted diesel, fewer wasted trips to the field.

DISCOM

Fewer subsidised units billed, higher solar self-consumption on MSKVY feeders.

STATE

Lower agricultural power subsidy against the same service commitment.

AQUIFER & AIR

Up to **750 ML** of groundwater left in the ground per feeder per kharif season, and **~91 t CO_2** avoided a year at India's grid emission factor.

One district proves it. The scheme replicates it.

11 Oct – 22 Nov 2026

Round 2 prototype

Replace FAO-56 stage coefficients with measured Sentinel-2 evapotranspiration, extend the backtest across districts and seasons, and ship the scheduler and dashboard. The Tier A model already runs.

Dec 2026 – Jan 2027

Finale & validation

Independent check of seasonal draft against CGWB observation wells; review with Schneider Electric mentors.

2027

District pilot with a DISCOM

Run in shadow mode alongside the existing rota on MSKVY feeders, then advisory, then live scheduling.

2028 onward

Replicate across segregated-feeder states

Gujarat, Karnataka, Madhya Pradesh, Rajasthan and Punjab already run the feeder architecture this depends on.

The limitations belong on a slide, not in a footnote.

Feeder-to-command-area mapping is not public.

We approximate it geographically from land use and transformer locations. DISCOMs hold the real mapping internally, so this is a deployment detail — not a design flaw.

Evapotranspiration is noisy on sub-hectare plots.

We aggregate to the feeder command area, where independent errors largely cancel. We do not claim plot-level accuracy, and the scheduler never needs it.

The behavioural saving is modelled, not measured.

Which is precisely why Tier A and Tier B are reported separately, and why the headline number is always the physical one.

The backtest is one district, one year, and has no satellite input yet.

Crop demand currently comes from FAO-56 stage coefficients, not measured Sentinel-2 evapotranspiration. Replacing that assumption with observation is exactly what Round 2 builds.

TEAM — to be confirmed on submission · KL University, Hyderabad